

Questions for an expert

Publishing, getting help, and how researchers think

A checklist for the call with Anthony. Not about the paper's details, but about how research and publishing work and how students can get help. ★ = ask first; □ = if there is time.

How to use this list

A 30-minute call has room for about **8–10 questions**. Pick the ★ ones that matter most to you, ask them in that order, and let him talk. Write down his exact words, names, links and numbers. Do not try to cover everything: leftover questions become a follow-up email.

1. The twelve to ask if time is short

- ★ Where would you submit this work, and what would make an editor say yes or no?
- ★ Would you post a preprint first? What could go wrong later with the journal?
- ★ How should two high-school students handle authorship, affiliations and acknowledgements, and should a mentor be listed?
- ★ What would a reviewer attack first?
- ★ If you were us, what is the smallest extra piece of work that would most improve the project?
- ★ How did you find your first mentors, and how should we ask researchers for help without being a burden?
- ★ Where can students realistically get computing time, data or analysis help?
- ★ What are the norms for disclosing AI tools used for code and writing?
- ★ How do you know when a project is finished, and when it needs more work?
- ★ How do you handle a negative result, and when do you pivot?
- ★ What do you wish you had known at our stage?
- ★ Would you look at a revised draft, and who else should we talk to?

2. Getting to know him

Start here; it builds rapport and tells you which of the later sections he can help with most.

- What do you work on, and what is the question you are most excited about right now?
- How did you get into research? What did your path look like when you were our age?
- What was your first paper like? Was it rejected, and what did you learn from it?
- What is the biggest mistake you have made in a project, and how did you find it?
Why: ours was a bug in our first version. Hearing his stories makes it easier to talk about yours honestly.
- What does a typical week look like: how much reading, experiments, writing, meetings?

3. Choosing where to publish

- ★ How do you decide where a paper belongs? What tells you a journal is a good fit?
Why: ask what he looks at: recent papers in the journal, article types, the editors, the scope statement.
- Journals also offer different article types (full papers, short notes, and so on). How do we know which one fits our work?
- Do benchmarking or negative-result papers get published, and where do they do well? Can you point to examples?
- Is it better to aim high and risk rejection, or to start with a realistic venue? How many rejections before you change the paper itself?

- ☐ How much does the journal's name matter for us, compared with getting the work out and read?
- ☐ How do conferences and workshops compare with journals in computational biology and machine learning, for value and timeline?
- ☐ Are there venues or programs designed for student research (student journals, symposia, poster sessions, science competitions)? Which are worth the effort and which are not?
- ☐ How do we spot a predatory or low-quality journal? Are there red flags or lists you trust?
- ☐ What are the most common reasons a paper is rejected without being sent to reviewers, and how do we avoid them?
- ☐ What do editors want to see in a cover letter? Is it worth suggesting reviewers, and how do we choose them?

4. How the process works

Preprints

- ★ **Should we post a preprint (for example on bioRxiv) before or after submitting? What are the benefits and risks?**

Why: you can check the target journal's preprint policy yourselves; ask about his experience, not the rules.

- ☐ Does a preprint help a student paper get noticed, or can it hurt?
- ☐ How do you handle comments or criticism that arrive on a preprint?

Authorship and credit

- ★ **How are authorship and author order decided? What counts as a contribution worth authorship, and what belongs in the acknowledgements?**

- ☐ When does a mentor become a co-author, and when is thanks enough? How should students raise this question politely?
- ☐ Who is the corresponding author, and does it matter? What email and affiliation should students use?
- ☐ Do reviewers or editors treat papers with student authors differently? Should we say we are students?

Honesty and disclosure

- ★ **What are the norms for disclosing AI tools that helped write code or text? How do you use them yourself, and what wording would you find acceptable?**

Why: ours received substantial help. Asking openly is the right move, and his answer tells you what wording to use.

- ★ **We had an earlier, flawed version of this work that was never submitted anywhere. Do we need to mention it? How should a paper describe a bug we found and fixed?**

- ☐ What should be public before submission: code, data, results? What is easy to forget (licence, archived copy, exact versions)?
- ☐ What would make a reviewer uncomfortable about a project like ours, and how can we address it up front?

Timelines, reviews and rejection

- ☐ How long does the process usually take from submission to a decision? What do you do while waiting? When is it fine to email the editor?
- ★ **How do you read a harsh review? How should we structure a response letter, and when is it right to push back?**
- ☐ After a rejection, what do you do first? How do you decide between resubmitting elsewhere and rethinking the work?
- ☐ How do you prioritise when reviewers ask for many extra experiments?
- ☐ How do students think about publication costs (open-access fees, waivers)? Who pays, and where do we find out before submitting?

5. Getting help

Mentors and collaborators

- ★ **How did you find your mentors? What makes a good one, and how would we find one?**
- ★ **How should students cold-email a professor or postdoc? What makes an email get a reply, what should it include, and how long should we wait before following up?**
- ☐ What can we reasonably ask a busy researcher for: a 15-minute chat, a look at a draft, a code review? What is too much?
- ☐ Is it appropriate to email the authors of the papers we build on? How would you word it?
- ☐ Do you know groups working on this kind of problem who might be open to hearing from students?
- ☐ How do we find a biologist to check our biological interpretation, and a statistician to check our analysis?

Resources

- ★ **Where can students get GPU or cluster time (university programs, cloud credits, paid notebooks, shared clusters)? Are any open to high-school students?**
- ☐ Are there summer research programs, lab-volunteer positions, or online communities (student groups of scientific societies, Discord or Slack groups) worth joining?
- ☐ Are there other datasets or benchmarks you would trust for this kind of work?
- ☐ Which tools, courses or books do you recommend for learning the statistics we are missing?

6. How researchers think

- ★ **How do you decide whether a result is real? What habits stop you from fooling yourself (controls, several random seeds, holding out data)?**
- ★ **How do you know when a project is done and it is time to write, versus needing another experiment?**
- ★ **How do you deal with a negative result, emotionally and professionally? How do you frame it honestly without underselling it?**
- ☐ What makes a good research question for a small team with limited compute?
- ☐ How do you catch bugs before they cost you months? What checks do you run (simple baselines, tests, sanity checks)?
- ☐ How do you keep experiments reproducible: notes, version control, saved settings?
- ☐ How do you balance rigour and speed? When is “good enough” good enough?
- ☐ How much is novelty valued in your field compared with careful replication and evaluation?
- ☐ How do you read papers efficiently? How many do you read in a week?
- ☐ What is your writing process: outline first, several drafts, who reads them?
- ☐ How do you ask for criticism, and how do you decide which criticism to act on?
- ☐ When a senior person disagrees with you, how do you handle it?

7. Education and career

- ★ **What do you wish you had known at our stage? What would you do differently in your early career?**
- ☐ If you were a high-school student today who wanted to do computational biology, what would you learn first: maths, programming, biology?
- ☐ What do you look for in a student researcher? What impresses you, and what puts you off?
- ☐ What routes lead into research as an undergraduate (majors, programs, labs)? How do students get into a

lab?

- ☐ How should we describe this project honestly on applications? What counts as impressive, and what looks inflated?
- ☐ For someone our age, does a publication matter, or depth of understanding? What matters more?
- ☐ How do people move between biology and machine learning? Which side is harder to learn later?
- ☐ Are there warning signs of bad research culture (hype, pressure to get results) we should avoid?
- ☐ How do you balance research with school and the rest of life?

8. Closing the call

- ★ **If you were us and had one month, what would you do?**
- ★ **Would you be willing to look at a revised draft in a few weeks? What format helps you most?**
- ☐ Who else should we talk to, and may we mention your name?
- ☐ What is one thing we did not ask that we should have?
- ☐ Would you like updates on how it goes?

9. Follow-up probes

Short prompts that turn an answer into something you can act on:

- “Can you say more about that?” · “What would you do in our shoes?”
- “What would change your mind?” · “Can you point us to an example?”
- “What is the smallest version of that we could do?” · “What is the most common mistake here?”
- “Is there something you would *not* do?” · “Who would know more about this?”

If he says...

“Submit somewhere smaller first.”	Which venue, and what does a strong submission there look like? Why is it a better fit?
“Get a mentor to look at it.”	How would we approach one? What would be a reasonable first request?
“Do more experiments.”	Which are essential and which are nice to have? What is the minimum to make it solid?
“Negative results are hard to publish.”	How do we frame it? Where have you seen them do well?
“Post a preprint.”	When, and what should we check about the journal’s policy first?
“AI use is fine if disclosed.”	Can you show us wording you would find acceptable?

10. Look these up yourselves first

Do not spend his time on things you can read. Check these before the call, and use the call for judgement and experience:

- ☐ The target journal’s instructions for authors: article types, length limits, formatting.
- ☐ The journal’s policies on preprints, on AI tools, and on authorship and contributions.
- ☐ Fees, open-access options, and whether waivers exist.
- ☐ Recent papers in the journal that resemble yours (evaluation or benchmarking studies).
- ☐ The data and code requirements: what must be shared and where.
- ☐ Anthony’s own work: a few of his papers, so your questions can refer to them.

Policies change and differ between journals; anything he tells you about them is worth confirming on the journal’s

own pages.

11. Manners that make experts want to help

- **Be specific.** “Should we post a preprint first?” gets a better answer than “any advice?”
- **Don’t ask for guarantees.** Not “will it be accepted?” but “what would make it stronger?”
- **Be honest** about what you did, including help from AI tools and the earlier bug. It builds trust.
- **Listen more than you talk.** Take notes; repeat back his two or three concrete suggestions at the end.
- **Respect his time.** Stay within the agreed length; offer to send questions in writing for anything left over.
- **Follow up.** Send a thank-you within a day listing what you will do, then a short progress update in three to four weeks. People help those who show progress.
- **Give back.** Offer to share results, code and anything you learn that might be useful to him.

Notes from the call